

▼ Titanic Survival Analysis

Maincrafts Technology Internship – Task 2

Objective

Analyze survival patterns in the Titanic dataset using Python and Data Visualization techniques.

Questions to Answer

- 1. Who survived more: males or females?
- 2. Did passenger class affect survival chances?
- 3. What was the survival rate by age group?

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set_style("whitegrid")
```

▼ Loading Titanic Dataset

```
df = pd.read_csv("train (1).csv")
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
				Futrelle, Mrs.								

Insight

The Titanic dataset contains passenger details such as age, gender, passenger class, fare, and survival status.

Dataset Information

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
print("Dataset Shape:", df.shape)
```

Dataset Shape: (891, 12)

```
df.columns
```

Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
 dtype='object')

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Insight

The dataset contains both numerical and categorical variables. Statistical summary helps understand the distribution of numerical features.

Missing Values Analysis

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Insight

Missing values are present in Age, Cabin, and Embarked columns. These must be handled before performing analysis.

Data Cleaning

```
df['Age'].fillna(df['Age'].median(), inplace=True)
```

/tmp/ipykernel_3606/1933487976.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=T

```
df['Age'].fillna(df['Age'].median(), inplace=True)
```

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

/tmp/ipykernel_3606/3744086084.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=T

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

```
df.drop('Cabin', axis=1, inplace=True)
```

```
df.isnull().sum()
```

```
      0
PassengerId  0
Survived     0
Pclass       0
Name         0
Sex          0
Age          0
SibSp        0
Parch        0
Ticket       0
Fare         0
Embarked     0
```

dtype: int64

Insight

Missing values were successfully handled. Age was filled using median values, Embarked was filled using mode, and Cabin was removed because it contained too many missing values.

✓ Survival Analysis by Gender

```
gender_survival = df.groupby('Sex')['Survived'].mean()

print(gender_survival)
```

```
Sex
female    0.742038
male      0.188908
Name: Survived, dtype: float64
```

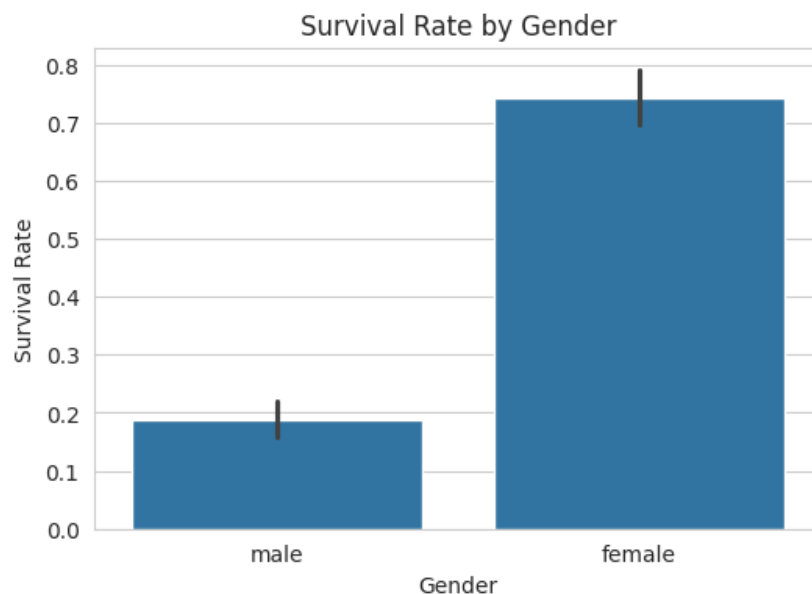
✓ Insight

The survival rate is significantly higher for females compared to males. This indicates that women had a greater chance of survival during the Titanic disaster.

```
plt.figure(figsize=(6,4))
sns.barplot(x='Sex', y='Survived', data=df)

plt.title("Survival Rate by Gender")
plt.xlabel("Gender")
plt.ylabel("Survival Rate")

plt.show()
```



Insight

The bar chart clearly shows that females had a much higher survival rate than males. This indicates that women were given priority during rescue operations.

✓ Survival Analysis by Passenger Class

```
class_survival = df.groupby('Pclass')['Survived'].mean()
print(class_survival)
```

```
Pclass
1    0.629630
2    0.472826
3    0.242363
Name: Survived, dtype: float64
```

✓ Insight

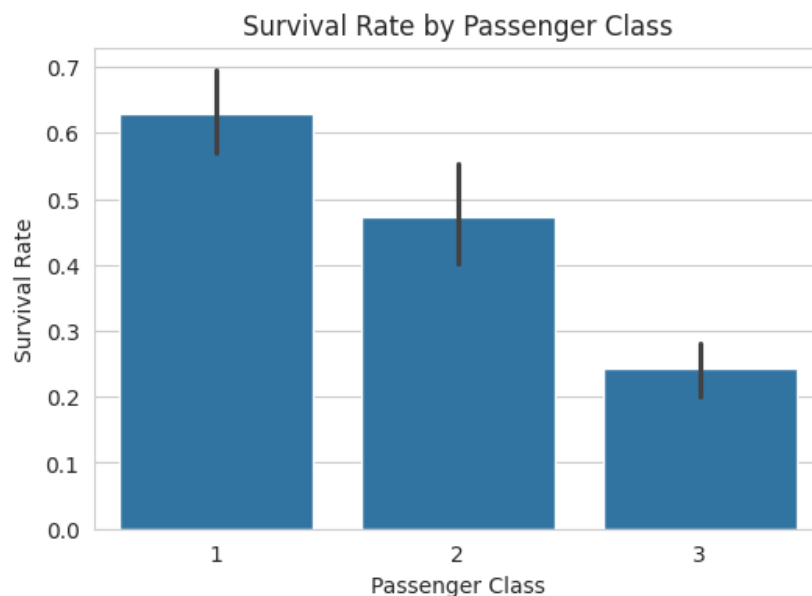
Passenger class appears to influence survival chances. First-class passengers had better access to lifeboats and rescue facilities.

```
plt.figure(figsize=(6,4))

sns.barplot(x='Pclass', y='Survived', data=df)

plt.title("Survival Rate by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Survival Rate")

plt.show()
```



Insight

Passengers travelling in First Class had the highest survival rate, while Third Class passengers had the lowest survival rate.

✓ Survival Analysis by Age Group

```
df['AgeGroup'] = pd.cut(
    df['Age'],
    bins=[0,12,18,35,60,100],
    labels=['Child', 'Teenager', 'Young Adult', 'Adult', 'Senior']
)
```

```
age_survival = df.groupby('AgeGroup')['Survived'].mean()

print(age_survival)
```

```
AgeGroup
Child      0.579710
Teenager   0.428571
Young Adult 0.353271
Adult      0.400000
Senior     0.227273
Name: Survived, dtype: float64
/tmp/ipykernel_3606/2998108859.py:1: FutureWarning: The default of observed=False is deprecated and will be
age_survival = df.groupby('AgeGroup')['Survived'].mean()
```

✓ Insight

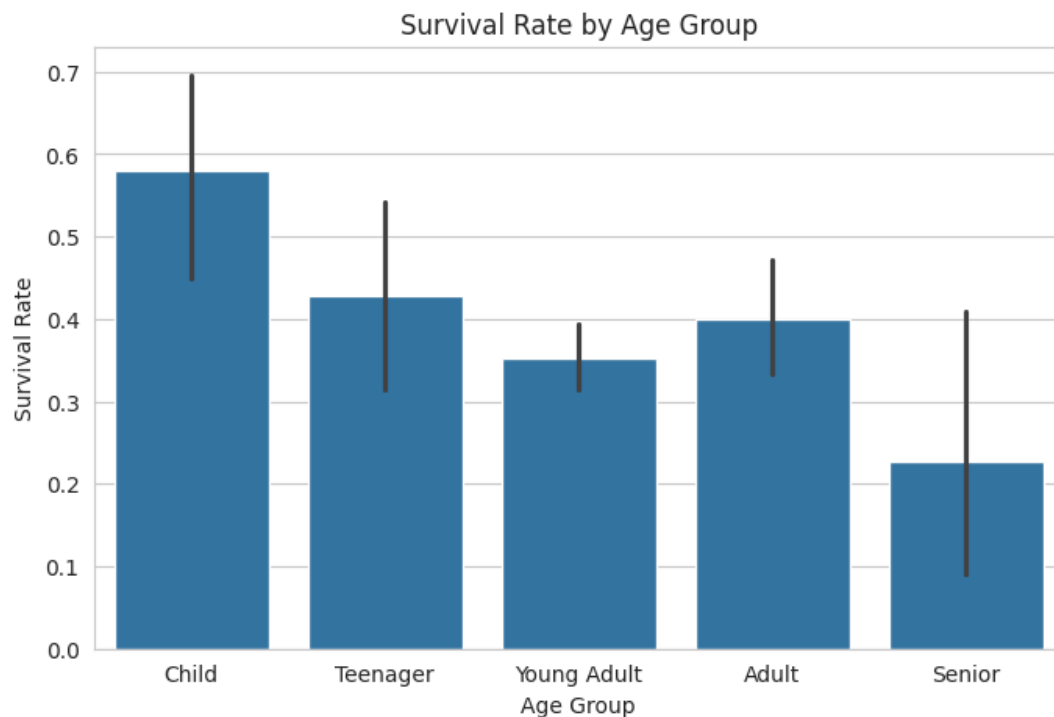
The survival rate varies across different age groups. Younger passengers generally had better survival chances compared to some older age categories.

```
plt.figure(figsize=(8,5))

sns.barplot(x='AgeGroup', y='Survived', data=df)
```

```
plt.title("Survival Rate by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Survival Rate")

plt.show()
```



Insight

The chart highlights how survival rates differ among age groups and helps identify which age category had the highest survival probability.

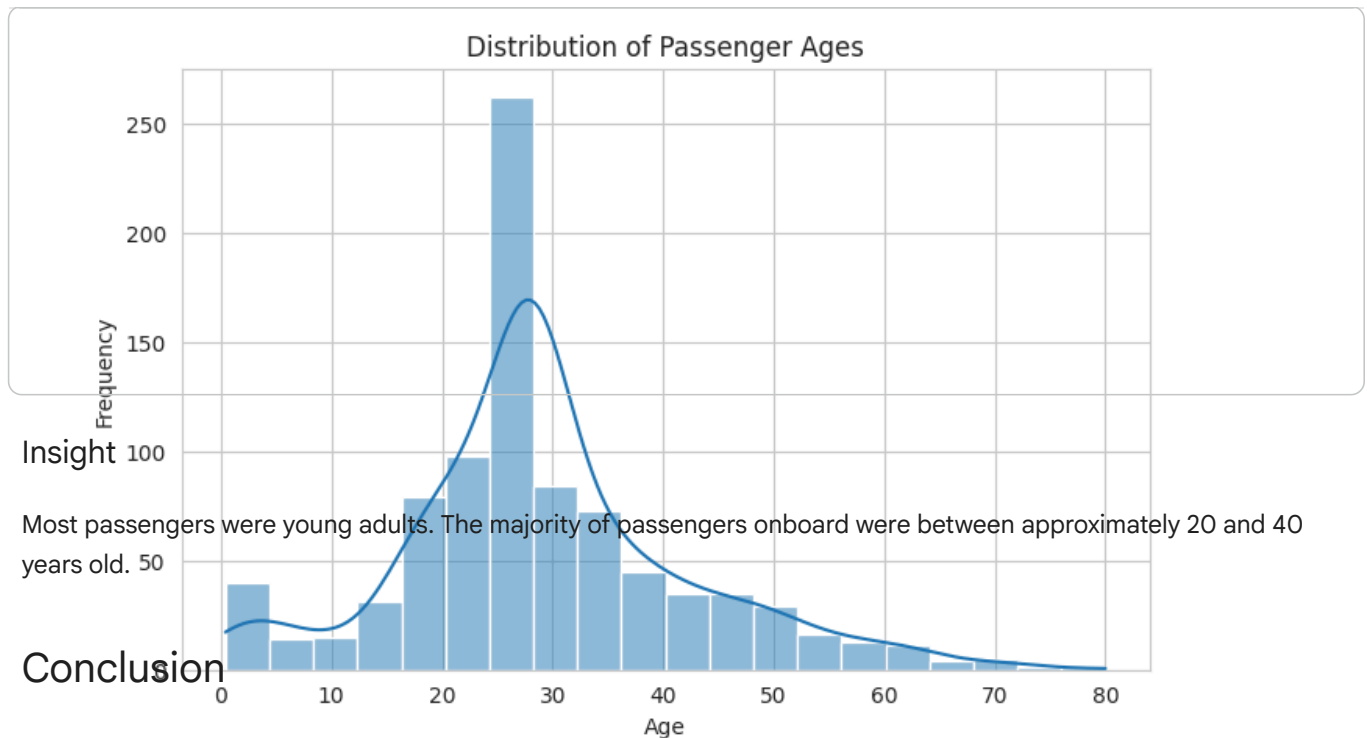
✓ Distribution of Passenger Ages

```
plt.figure(figsize=(8,5))

sns.histplot(df['Age'], bins=20, kde=True)

plt.title("Distribution of Passenger Ages")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```



This project analyzed the Titanic dataset to understand survival patterns among passengers.

Key Findings

- Females had a significantly higher survival rate than males.
- Passenger class strongly influenced survival chances.
- Different age groups showed different survival patterns.
- Data visualization helped uncover meaningful trends and relationships.

Overall Outcome

This project successfully demonstrated:

- Data Cleaning
- Missing Value Handling
- Exploratory Data Analysis (EDA)
- Data Visualization
- Survival Pattern Analysis
- Insight Generation

The analysis provided valuable understanding of the factors that influenced passenger survival during the Titanic disaster.